

PLAN & FETCH · r0-11

- [plan] LLaMA-130M: Pre-LN perplexity 26.73 vs Pre-LN+LNS 25.76 r0
- \$ git clone https://github.com/lmsdss/LayerNorm-Scaling repo r5

BUILD · r12-18

- \$ uv pip install ... transformers==4.44.2 tokenizers==0.19.1 # aarch64 wheels r17

DEBUG & PATCH · r19-26

- \$ NORM_TYPE=LNS POST_NUM=3 ../.venv/bin/python -c ... LlamaForCausalLM(cfg) r19
- UnboundLocalError: cannot access local variable 'present_key_value' r19
- \$ diff /repro/reference/supplement/... peft_pretraining/modeling_llama.py r21
- 427c427 ... elif norm_type == 'cod': ... elif norm_type == 'LNS': r21
- `elif norm\_type == 'LNS'` is DEAD CODE (since 'lns' != 'LNS') decisive r22
- \$ edit_file ... replace_all for `norm\_type == 'LNS'` → `norm\_type == 'lns'` r25
- max abs diff LNS vs PRE: 1.8720629215240479 r26

RUN & TRAIN · r27-45

- \$ NORM_TYPE=LNS ... torchrun_main.py ... --num_training_steps 20000 # 3.03 GPU-hours GPU r41
- \$ NORM_TYPE=pre ... torchrun_main.py ... --run_name "130m_res_pre_lr0.001" GPU r44
- LNS: PPL = 24.93 (paper: 25.76) ... Pre-LN: PPL = 25.07 (paper: 26.73) r45

VERIFY & REPORT · r46-51

- \$ write_file evidence/REPORT.md r47

lns ppl 24.93 vs paper 25.76; pre-ln 25.07 vs 26.73 audit 8, reproduced ✓